AI/ML INTERNSHIP - DS GROUP

Progress Report: NLP Sentiment Modeling

INTERN	PERIOD	REPORT DATE	STATUS
[Ishaan Sharma]	Weeks 1–3 (Days 1–21)	Sept 24, 2026	On Schedule

Executive Summary

Week 3 Complete

Full baseline Machine Learning pipeline delivered with end-to-end data cleaning, hyperparameter optimization, and empirical transformer benchmark comparisons on the Amazon Fine Food Reviews dataset.

RETAINED DATA	BEST MODEL	TEST ACCURACY	MACRO F1
393,890	**Linear SVM**	**95.61%**	**0.9141**
From 568k clean rows	TF-IDF (1-3 N-grams)	+0.19% vs baseline	+0.0032 post-tuning

Week 1: Data Onboarding & EDA

Dataset & Cleaning Pipeline

- **Source Data:** Kaggle `snap/amazon-fine-food-reviews`.
- **Deduplication:** Removed 174,562 rows sharing identical `UserId + Time + Text`.
- **Data Quality:** Dropped 2 broken rows with corrupted helpfulness values.
- **Retention Rate:** Retained 69.3% of raw volume (393,890 clean reviews).

Feature Engineering & EDA

- **Target Variable:** `sentiment_label` (Score 4-5 = Positive, 1-2 = Negative, 3 = Neutral).
- **Temporal Attributes:** Derived `review_date`, `year`, and `month`.
- **Metrics Built:** Text lengths & `helpfulness_ratio`.
- **Class Split:** 77.95% Positive, 14.50% Negative, 7.56% Neutral.

Mentor Feedback Applied

1. Dropped redundant metadata columns (`Id`, `ProfileName`, `Time`, raw counts).
2. Implemented Word Clouds for `Summary` breakdowns & per-column descriptive statistics.
3. Built stratified cross-validation framework for robust model evaluations.

Week 2: NLP Exploration & Feature Intelligence

- **Text Processing:** Built pipeline featuring lower casing, punctuation stripping, stopword removal.
- **N-gram Extraction:** Processed 1-3 word phrase patterns using `CountVectorizer(ngram_range=(1,3), min_df=5)`.
- **Business Queries:** Formulated 16 SQL/Pandas analytics queries to track rating movements, length correlations, and seller trends.

Week 2: Critical Analytical Insight

> **Helpfulness Correlation vs. Artifact Discovery**
>
> **Initial Hypothesis:** Positive correlation observed between review length and helpfulness ratio.
> **Investigation & Quality Issue:** Reviews with zero votes (0/0) were creating undefined ratios. Excluding zero-vote reviews revealed that length increases vote volume, but not ratio quality.
> **Analytical Implication:** Disproved naive correlation. Undefined ratios are treated as missing data, preventing false signal propagation in auxiliary tasks.

Week 3: Baseline ML & Hyperparameter Tuning

Model Strategy: Binary classification (Positive vs. Negative, 364,122 reviews). Stratified 80/20 train/test split (291,297 train / 72,825 held-out test). Held-out test set was strictly untouched during training and tuning. Implemented 2-fold stratified cross-validation in GridSearchCV as a memory-safe strategy for 16GB RAM constraints.

Baseline Classification Performance

Model	Accuracy	Macro F1	Neg F1	Pos F1
Linear SVM (Default)	**95.42%**	**0.9109**	**0.8487**	**0.9730**
Logistic Regression (Default)	95.21%	0.9040	0.8361	0.9718
Linear SVM (Balanced)	94.07%	0.8960	0.8279	0.9641
Logistic Regression (Balanced)	93.87%	0.8943	0.8257	0.9629
Naive Bayes	92.11%	0.8210	0.6871	0.9548
Majority Baseline	84.32%	0.4575	0.0000	0.9150

GridSearchCV Tuning Configuration

```
      param_grid = { 'clf__C': [0.1, 0.5, 1.0], 'tfidf__max_features': [40000, 50000] } # Fixed
Pipeline: TfidfVectorizer(ngram_range=(1,3), min_df=5) -> LinearSVC()
```

Optimized Parameters & Rationale

- **Best C (0.5):** Provided slight regularization smoothing over default C=1.0.
- **Vocab Limit (40,000):** Expanded from 20k to maximize n-gram coverage.
- **N-Gram Range (1, 3):** Trigrams captured critical context like "not worth the money".
- **Training Time:** 7.1 mins (12 fits: 6 combinations × 2 folds).

Key Architectural Decisions

- **Default vs. Balanced Weights:** Kept default weights because class balancing dropped overall accuracy (95.42% → 94.07%) while negative F1 remained strong (0.85).
- **Inference Batch Size (64):** Configured for transformer inference GPU/RAM efficiency.
- **Test Set Isolation:** Held out 72,825 samples untouched during tuning.

Zero-Shot Transformer Benchmarking

Evaluated off-the-shelf pre-trained transformer models on the held-out test set to determine baseline neural performance against classical approaches without fine-tuning.

Model Architecture	Accuracy	Macro F1	Domain / Pre-training
Linear SVM (TF-IDF Tuned)	**95.61%**	**0.9141**	**In-domain Fine-grained Baseline**
VADER (Lexicon)	88.51%	0.7413	Rule-based Sentiment Rules
RoBERTa (Twitter Zero-Shot)	85.94%	0.7910	Social Media General Domain
NLP Town BERT (Zero-Shot)	78.11%	0.7164	Product/Yelp Multilingual
DistilBERT (SST-2 Zero-Shot)	67.82%	0.6235	Movie Reviews Binary

Benchmark Takeaway

Classical ML (TF-IDF + Linear SVM) outperformed zero-shot transformers by **10+ F1 points**. Off-the-shelf transformers struggle with domain shift and e-commerce vocabulary without explicit fine-tuning.

Decision Rationale Matrix

Decision Area	Option Chosen	Engineering & Analytical Justification
Data Cleaning	Exclude 0/0 Vote Rows	Prevents false correlation artifact where longer text appears more helpful due to raw vote volume alone.
Class Weighting	Default (Unweighted)	Class-balanced weighting degraded overall accuracy (94.07% vs 95.42%) without improving negative F1.
CV Strategy	2-Fold Stratified CV	Optimized cross-validation pipeline to guarantee memory safety within 16GB RAM on 291k samples.
Feature Space	Trigrams + 40k Vocab	Captures key contextual phrases ("not worth the money") yielding a +0.32 F1 point lift.

Error Analysis & Qualitative Findings

False Positives (n=2,053 / 2.82%)

Predicted Positive, True Negative:

- Mixed sentiment ("tastes great... but goes bad quickly").
- Sarcastic phrasing ("lovely if you enjoy cardboard").

False Negatives (n=1,286 / 1.77%)

Predicted Negative, True Positive:

- Neutral wording in positive context.
- Implicit praise ("not fancy, just good coffee").

Next Steps (Weeks 4–6 Roadmap)

- **Transformer Fine-Tuning:** Domain-adapt DistilBERT / DeBERTa on 290k training subset using Hugging Face Trainer.
- **Ensembling:** Blending Linear SVM decision function outputs with Transformer logits